TECHNICAL DILIGENCE REVIEW - RECORD 003

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-003

1. Engagement scope, assessors and evidence conventions

Ardent Peak commissioned the report at TDR-2025-11 from Wrenfield Technical Advisory LLP to test the security architecture, penetration-test findings and vulnerability management of the platform ahead of the Series B financing. This report, report TDR-2025-11, is the Wrenfield Technical Advisory LLP workstream on the security architecture, penetration-test findings and vulnerability management of the platform, delivered to Ardent Peak on 2025-09-23. 5 teams contributed to the interview programme for TDR-2025-11, and one of them was interviewed twice after a document arrived late. 18 interviews inform the report at TDR-2025-11. Where an interview is the sole source for a claim, the claim is graded limited. The reader of this report should treat each conclusion as bounded by the section it appears in.

The present report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. Work on the present report ran over 3 weeks and included 18 interviews and read-only access to the engineering repositories. 3 items in the report at TDR-2025-11 are post-close engineering work rather than transaction issues, and the reviewer has costed them at $1,280,000. The report at TDR-2025-11 recommends that 1 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. J. Ferreiro reports management's own position wherever it differs from the reviewer's, and does not merge the two.

Grading in the present report is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. J. Ferreiro treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. J. Ferreiro did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. Nothing in the report at TDR-2025-11 rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. J. Ferreiro has resisted rounding the findings towards a comfortable conclusion in either direction.

The sample behind the report at TDR-2025-11 is 20 services and 7 sites; the selection was the reviewer's, not the company's. For this report the reviewer sampled 7 sites, including the largest deployment and the newest. 8 items in report TDR-2025-11 are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. 8 matters remain open at the date of TDR-2025-11, and none of them is closed by the passage of time alone. J. Ferreiro has separated what the company has built from what the company can demonstrate about what it has built.

The principal limitation on report TDR-2025-11 is time: 3 weeks was enough to test the control path and not enough to test the reporting path. Two document requests remained partly unsatisfied when TDR-2025-11 was delivered, and the affected conclusions are marked. 2025-09-23 is the evidence cut-off for report TDR-2025-11, whose subject is the security architecture, penetration-test findings and vulnerability management of the platform. The instruction behind this report was narrow: examine the security architecture, penetration-test findings and vulnerability management of the platform and report by 2025-09-23. J. Ferreiro has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

Interviewees for TDR-2025-11 were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. Report TDR-2025-11 draws on 18 interviews, including the engineering owners of every component examined in detail. Selection for TDR-2025-11 favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. Coverage of report TDR-2025-11 is deliberately uneven: the safety and control path was examined closely and the reporting path only at a high level. Where TDR-2025-11 uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-11)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-03-001	Fleet Allocator: test report	test report	2025-05-24	the safety controller vendor	corroborated an incident narrative
EV-03-002	Speed and Separation Monitor: configuration export	configuration export	2025-05-19	the sensor vendor	established a dependency

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-03-003	Fleet Allocator: design record	design record	2025-12-29	the safety controller vendor	supported a coverage figure
EV-03-004	dependency and licence scan: incident post-mortem	incident post-mortem	2025-09-26	Helioscope field reliability	evidenced an approval
EV-03-005	Robot Shadow Store: incident post-mortem	incident post-mortem	2025-10-25	Helioscope safety	established a dependency
EV-03-006	Analytics Warehouse Loader: measurement extract	measurement extract	2026-03-26	Helioscope field reliability	evidenced a remediation
EV-03-007	Site Provisioning Service: policy document	policy document	2026-01-26	Helioscope security	established a dependency
EV-03-008	Lidar Driver: architecture note	architecture note	2025-11-03	the sensor vendor	verified a stated objective
EV-03-009	Payload Scale Interface: runbook	runbook	2026-02-10	Helioscope safety	established a dependency
EV-03-010	hardware-in-the-loop, bench: release record	release record	2026-06-19	the sensor vendor	established a dependency
EV-03-011	Event Bus: incident post-mortem	incident post-mortem	2025-11-07	the sensor vendor	evidenced an approval
EV-03-012	integration tests, cloud: incident post-mortem	incident post-mortem	2025-07-24	Redwater Security	verified a stated objective
EV-03-013	Costmap Fusion: design record	design record	2025-10-09	Redwater Security	verified a stated objective
EV-03-014	Alarm Router: audit output	audit output	2026-09-08	Meridian Assurance	corroborated an incident narrative
EV-03-015	Safety Controller: release record	release record	2025-11-28	Helioscope safety	evidenced a remediation
EV-03-016	Billing Usage Meter: runbook	runbook	2025-08-13	Helioscope safety	supported a coverage figure
EV-03-017	performance smoke, cloud: measurement extract	measurement extract	2025-08-05	Helioscope safety	evidenced an approval
EV-03-018	Analytics Warehouse Loader: measurement extract	measurement extract	2026-07-13	Meridian Assurance	evidenced an approval
EV-03-019	container image vulnerability scan: audit output	audit output	2026-06-07	Helioscope field reliability	verified a stated objective
EV-03-020	unit tests, cloud services: policy document	policy document	2025-08-17	Helioscope field reliability	corroborated an incident narrative
EV-03-021	Safety Controller: audit output	audit output	2025-10-28	Meridian Assurance	supported a capacity conclusion
EV-03-022	Gripper Controller: vendor statement	vendor statement	2026-07-26	the sensor vendor	supported a coverage figure
EV-03-023	Alarm Router: runbook	runbook	2026-10-13	the sensor vendor	established a dependency
EV-03-024	Traffic Manager: vendor statement	vendor statement	2026-01-03	Redwater Security	evidenced a remediation
EV-03-025	build, robot firmware images: runbook	runbook	2026-04-29	Helioscope safety	corroborated an incident narrative
EV-03-026	Alarm Router: configuration export	configuration export	2026-08-01	Meridian Assurance	supported a coverage figure
EV-03-027	simulation suite, nightly: audit output	audit output	2025-06-09	Helioscope safety	verified a stated objective
EV-03-028	Alarm Router: vendor statement	vendor statement	2025-05-10	the sensor vendor	evidenced a remediation
EV-03-029	Motion Planner: interview minute	interview minute	2026-05-23	Helioscope security	evidenced an approval
EV-03-030	dependency and licence scan: runbook	runbook	2025-12-08	the safety controller vendor	supported a coverage figure
EV-03-031	Release Orchestrator: configuration export	configuration export	2025-07-10	Helioscope engineering	established provenance
EV-03-032	Analytics Warehouse Loader: policy document	policy document	2026-04-24	Helioscope engineering	established a dependency

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-03-033	E-Stop Chain: runbook	runbook	2026-05-28	the sensor vendor	established a dependency
EV-03-034	Gripper Controller: design record	design record	2026-10-02	Meridian Assurance	established a dependency
EV-03-035	performance smoke, cloud: vendor statement	vendor statement	2025-12-20	the sensor vendor	verified a stated objective
EV-03-036	Safety Controller: measurement extract	measurement extract	2026-08-03	Helioscope safety	verified a stated objective

1. Engagement scope, assessors and evidence conventions - part 2 of 2

For length, the 1. Engagement scope, assessors and evidence conventions is presented in 2 parts; this is the next of them, and the rows continue the same sequence. No row is omitted at the join.

Nothing in this part varies how the 1. Engagement scope, assessors and evidence conventions was compiled, and the headings given at the start remain the operative ones.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-11) (continued, part 2 of 2)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-03-037	Charge Scheduler: incident post-mortem	incident post-mortem	2025-07-23	the sensor vendor	established a dependency
EV-03-038	container image vulnerability scan: release record	release record	2026-10-28	Redwater Security	evidenced a remediation
EV-03-039	E-Stop Chain: test report	test report	2026-11-07	the safety controller vendor	corroborated an incident narrative
EV-03-040	contract tests, warehouse connectors: configuration export	configuration export	2026-08-17	Helioscope engineering	supported a coverage figure
EV-03-041	Simulation Harness: interview minute	interview minute	2025-06-11	Helioscope safety	established a dependency
EV-03-042	Payload Scale Interface: release record	release record	2026-10-04	Meridian Assurance	corroborated an incident narrative
EV-03-043	Fleet Health Monitor: release record	release record	2026-11-08	Helioscope safety	supported a capacity conclusion
EV-03-044	Battery Management Service: audit output	audit output	2026-05-28	the sensor vendor	established a dependency
EV-03-045	performance smoke, cloud: test report	test report	2025-05-22	the safety controller vendor	evidenced a remediation
EV-03-046	Motion Planner: vendor statement	vendor statement	2025-09-23	Redwater Security	evidenced a remediation
EV-03-047	Simulation Harness: runbook	runbook	2025-12-28	Redwater Security	evidenced a remediation
EV-03-048	Video Snippet Service: test report	test report	2026-05-17	Helioscope field reliability	evidenced an approval
EV-03-049	secret scan: release record	release record	2026-06-08	Redwater Security	established provenance
EV-03-050	WMS Connector: design record	design record	2026-02-15	Helioscope engineering	established a dependency
EV-03-051	Telemetry Ingest: policy document	policy document	2025-11-06	Helioscope security	supported a coverage figure
EV-03-052	WMS Connector: policy document	policy document	2025-09-17	the sensor vendor	corroborated an incident narrative
EV-03-053	Costmap Fusion: incident post-mortem	incident post-mortem	2026-01-21	Helioscope security	verified a stated objective
EV-03-054	container image vulnerability scan: test report	test report	2026-05-19	Helioscope safety	established a dependency
EV-03-055	Localization Engine: configuration export	configuration export	2026-03-13	the safety controller vendor	evidenced a remediation
EV-03-056	container image vulnerability scan: configuration export	configuration export	2026-07-25	Redwater Security	established a dependency
EV-03-057	Fleet Health Monitor:	architecture note	2025-12-16	the sensor vendor	supported a capacity

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	architecture note				conclusion
EV-03-058	unit tests, robot software: test report	test report	2025-10-21	the safety controller vendor	evidenced an approval
EV-03-059	static analysis, robot software: runbook	runbook	2026-05-03	Helioscope safety	corroborated an incident narrative
EV-03-060	Payload Scale Interface: release record	release record	2026-03-24	the sensor vendor	evidenced a remediation
EV-03-061	container image vulnerability scan: design record	design record	2026-06-05	Helioscope field reliability	supported a coverage figure
EV-03-062	Simulation Harness: configuration export	configuration export	2026-01-18	Redwater Security	established provenance
EV-03-063	build, cloud container images: interview minute	interview minute	2026-05-20	Helioscope field reliability	verified a stated objective
EV-03-064	WMS Connector: runbook	runbook	2026-02-02	Redwater Security	evidenced a remediation
EV-03-065	Billing Usage Meter: interview minute	interview minute	2025-12-18	the safety controller vendor	corroborated an incident narrative
EV-03-066	Telemetry Ingest: incident post-mortem	incident post-mortem	2026-01-13	Helioscope engineering	established a dependency
EV-03-067	Remote Operations Console: measurement extract	measurement extract	2025-11-29	the sensor vendor	corroborated an incident narrative
EV-03-068	Identity and Authorization: vendor statement	vendor statement	2026-06-01	Helioscope safety	established provenance
EV-03-069	Site Provisioning Service: measurement extract	measurement extract	2025-12-05	the sensor vendor	established a dependency
EV-03-070	Fleet Allocator: audit output	audit output	2025-05-15	Helioscope security	evidenced a remediation
EV-03-071	Battery Management Service: vendor statement	vendor statement	2026-10-20	the safety controller vendor	evidenced a remediation
EV-03-072	simulation suite, short: architecture note	architecture note	2025-05-05	the sensor vendor	verified a stated objective

2. SEC-2026-03 and the high-severity findings

Assessment finding SEC-2026-03 was opened on 2026-04-15 and states that config bundle signature verification could be disabled by a local flag. The reviewer's position on SEC-2026-03 is that the risk acceptance lacks an expiry date, which is a process finding in its own right. Assessment finding SEC-2026-03 translates into the following exposure: a local compromise on a robot becomes a persistent configuration compromise. Residual risk after remediation of Config Distribution finding SEC-2026-03 is low; the flag was removed and the agent refuses unsigned bundles. finding SEC-2026-03 is included because a reader of TDR-2025-11 should be able to disagree with the inference without repeating the work.

The tester scored Video Snippet Service finding SEC-2025-06 at 6.5 and labelled it Medium. Moderate evidence supports SEC-2025-06 against Video Snippet Service, namely that enumeration confirmed; no clip content was retrievable. Reading SEC-2025-06 against Video Snippet Service, the defect is that video snippet index allowed cross-tenant enumeration by identifier; the discovery date is 2025-09-19. 2025-12-18 is the closure position on SEC-2025-06 against Video Snippet Service, which the ledger records as remediated. assessment finding SEC-2025-06 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Identity and Authorization finding SEC-2026-08 is remediated, with a closure date of 2026-06-30. For SEC-2026-08 against Identity and Authorization the record shows that role grant performed in a staging tenant, which is strong evidence and is graded separately from severity. Severity on SEC-2026-08 is Medium, scored 6.4 by the assessor. Residual risk after remediation of finding SEC-2026-08 is low; role grants now require a second approver and are alerted. Medium-severity finding SEC-2026-08 carries a follow-up request that was satisfied after the draft of TDR-2025-11 was circulated.

Finding SEC-2026-07 was graded Medium, band 6.7, by Meridian Assurance engagement MA-2026-1. The finding at SEC-2026-07, dated 2026-04-17, is that hardware-in-the-loop rig controller runs an unsupported operating system. Assessment finding SEC-2026-07 is open, with a closure date of committed 2026-10-31. What remains after SEC-2026-07 against Hardware-in-the-Loop Rig is remediated is medium; network segmentation is the only current control. SEC-2026-07 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-11.

2025-09-22 is the discovery date of Site Provisioning Service finding SEC-2025-07, which finds that site provisioning stored a shared bootstrap secret in the template. Finding SEC-2025-07 is not fully retired by its remediation: medium; two forked templates were not re-issued until May 2026. Remediation status on SEC-2025-07 is remediated; the record shows 2026-01-16. SEC-2025-07 against Site Provisioning Service matters because the same secret was reusable at any site provisioned from that template. Medium-severity finding SEC-2025-07 was raised again with the owning engineer after the initial interview, and the position did not change.

SEC-2025-09 is supported by the fact that key custody was evidenced by interview and the key ceremony record; evidence strength is moderate. Finding SEC-2025-09 leaves residual risk that the reviewer assesses as medium; a two-person rule is designed but not enforced in the tool. Assessment finding SEC-2025-09 records that oTA artefact signing keys held by three engineers without a quorum control, discovered on 2025-09-24. The tester scored Medium-severity finding SEC-2025-09 at 6.3 and labelled it Medium. SEC-2025-09 is the clearest instance in TDR-2025-11 of engineering practice being better than the documentation suggests.

As at the review date SEC-2026-18 was open and the closure entry reads committed 2026-12-15. Read alongside the remediation record, SEC-2026-18 against Battery Management Service shows that the risk acceptance lacks an expiry date, which is a process finding in its own right. What Battery Management Service finding SEC-2026-18 enables is this: a diagnostic flood can slow pack telemetry reporting. On Battery Management Service finding SEC-2026-18 the tester's evidence was that flood produced on a bench pack, and the reviewer grades it limited, resting on a configuration read rather than on a demonstrated exploitation. assessment finding SEC-2026-18 rests on a single source, and the reviewer has said so rather than implying corroboration.

Rating for Medium-severity finding SEC-2026-16 is Medium, band 5.8; the reviewer re-read the scoring rationale and accepted it. 2026-04-22 is the discovery date of SEC-2026-16, which finds that costmap fusion diagnostic socket bound to all interfaces. Medium-severity finding SEC-2026-16 stands remediated; 2026-06-26 is what the record gives for closure. What SEC-2026-16 enables is this: on-network access to a live perception diagnostic stream. finding SEC-2026-16 is the reason one section of TDR-2025-11 concludes narrowly rather than generally.

Traffic Manager finding SEC-2026-14 is rated Low with a numeric band of 3.8 on the tester's scoring scale. Tracking Traffic Manager finding SEC-2026-14 to closure gives a status of remediated and a date of 2026-06-18. What remains after SEC-2026-14 against Traffic Manager is remediated is low. Impact of finding SEC-2026-14 is not theoretical; a manual corridor override cannot be attributed after the fact. assessment finding SEC-2026-14 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Finding SEC-2026-15 reduces to a residual risk of low; management judged the operational cost of enforcement higher than the risk. Assessment finding SEC-2026-15 matters because operator confusion rather than a direct security impact. Assessment finding SEC-2026-15 is accepted, with a closure date of not planned. Severity on SEC-2026-15 against Site Provisioning Service is Low, scored 3.0 by the assessor. Low-severity finding SEC-2026-15 is one of the entries where TDR-2025-11 relies on the company's own measurement rather than an independent one.

Schedule K - Penetration-test findings by severity with residual risk (TDR-2025-11)

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-01	High	7.4	Identity and Authorization	2025-09-16	2025-11-14	low once lifetimes are cut to one hour with refresh
SEC-2025-02	High	8.1	Release Orchestrator	2025-09-17	2025-09-19	low; redaction is enforced in the logging library
SEC-2025-03	Medium	6.2	Remote Operations Console	2025-09-18	2026-01-22	low; pinning is applied at the gateway and the agent
SEC-2025-04	Medium	5.4	Map Authoring	2025-09-18	2025-11-27	medium; the

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
			Studio			shortened publication path is still outstanding
SEC-2025-05	High	7.8	Telemetry Ingest	2025-09-19	2026-02-06	low; attestation is now required on the ingest path
SEC-2025-06	Medium	6.5	Video Snippet Service	2025-09-19	2025-12-18	low; the index is scoped per tenant and the identifiers are opaque
SEC-2025-07	Medium	5.9	Site Provisioning Service	2025-09-22	2026-01-16	medium; two forked templates were not re-issued until May 2026
SEC-2025-08	Medium	6.0	Event Bus	2025-09-23	2025-10-31	low; the interface is restricted to the operations subnet
SEC-2025-09	Medium	6.3	OTA Updater	2025-09-24	pending	medium; a two-person rule is designed but not enforced in the tool
SEC-2025-10	Low	3.7	Charge Scheduler	2025-09-24	2025-10-09	low
SEC-2025-11	Low	3.1	Remote Operations Console	2025-09-25	2025-10-30	low
SEC-2025-12	Low	4.2	Analytics Warehouse Loader	2025-09-25	not planned	medium; management accepted the risk with a compensating audit trail
SEC-2026-01	High	7.1	Release Orchestrator	2026-04-14	committed 2026-08-31	high until the second-approver control lands
SEC-2026-02	High	7.6	Robot Shadow Store	2026-04-15	2026-05-29	low; authorization is checked on both the cache and durable paths
SEC-2026-03	High	8.4	Config Distribution	2026-04-15	2026-06-12	low; the flag was removed and the agent refuses unsigned bundles
SEC-2026-04	Medium	5.6	WMS Connector	2026-04-16	2026-05-20	low; payload logging requires an explicit per-tenant approval
SEC-2026-05	Medium	5.2	Alarm Router	2026-04-16	committed 2026-09-30	medium; per-region credentials are scheduled but not issued
SEC-2026-06	Medium	6.1	Simulation Harness	2026-04-17	pending	medium; the role was scoped down but still reads one production bucket
SEC-2026-07	Medium	6.7	Hardware-in-the-Loop Rig	2026-04-17	committed 2026-10-31	medium; network segmentation is the only current control
SEC-2026-08	Medium	6.4	Identity and Authorization	2026-04-18	2026-06-30	low; role grants now require a second approver and are alerted
SEC-2026-09	Low	3.9	Fleet Health Monitor	2026-04-18	2026-05-08	low
SEC-2026-10	Low	4.0	Billing Usage Meter	2026-04-19	2026-06-05	low; object lock is applied and verified nightly
SEC-2026-11	Low	4.4	Gripper Controller	2026-04-19	committed 2026-11-30	medium; the safety controller limit is the

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
						compensating control
SEC-2026-12	Low	3.5	Edge Gateway Agent	2026-04-20	pending	medium; a wipe procedure exists but was not applied retroactively
SEC-2026-13	Low	3.3	Order Broker	2026-04-20	2026-05-15	low
SEC-2026-14	Low	3.8	Traffic Manager	2026-04-21	2026-06-18	low
SEC-2026-15	Low	3.0	Site Provisioning Service	2026-04-21	not planned	low; management judged the operational cost of enforcement higher than the risk
SEC-2026-16	Medium	5.8	Costmap Fusion	2026-04-22	2026-06-26	low; the socket binds to the loopback interface only
SEC-2026-17	High	7.9	Motion Planner	2026-04-22	2026-06-19	low; the port requires a signed local credential
SEC-2026-18	Low	3.4	Battery Management Service	2026-04-23	committed 2026-12-15	low; no fleet impact demonstrated
SEC-2026-19	Medium	5.1	Localization Engine	2026-04-23	committed 2026-10-15	medium; disk encryption is planned for the next compute revision
SEC-2026-20	Low	3.2	Payload Scale Interface	2026-04-24	not planned	low; management accepted pending the calibration service redesign

3. Medium and low findings, ageing and residual risk

Assessment finding SEC-2025-05 is rated High with a numeric band of 7.8 on the tester's scoring scale. As at the review date assessment finding SEC-2025-05 was remediated and the closure entry reads 2026-02-06. After the fix, SEC-2025-05 against Telemetry Ingest carries residual risk described as low; attestation is now required on the ingest path. On High-severity finding SEC-2025-05 the conclusion is that the risk acceptance lacks an expiry date, which is a process finding in its own right. finding SEC-2025-05 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Residual risk after remediation of SEC-2026-01 against Release Orchestrator is high until the second-approver control lands. Limited, resting on a configuration read rather than on a demonstrated exploitation evidence supports SEC-2026-01 against Release Orchestrator, namely that the tester used the path in a staging tenant and the audit trail recorded no justification. Weighing High-severity finding SEC-2026-01 against the compensating controls, the underlying pattern appears in two other components that the tester did not reach. Emergency release path permits a single approver is the substance of SEC-2026-01, raised on 2026-04-14. SEC-2026-01 appears twice in the working papers for TDR-2025-11 because the first answer was corrected after a follow-up request.

What finding SEC-2026-19 enables is this: a stolen compute module discloses a customer's warehouse layout. Tracking Medium-severity finding SEC-2026-19 to closure gives a status of open and a date of committed 2026-10-15. On SEC-2026-19 against Localization Engine the tester's evidence was that cache read from a removed module, and the reviewer grades it moderate. Finding SEC-2026-19 was graded Medium, band 5.1, by Meridian Assurance engagement MA-2026-1. Localization Engine finding SEC-2026-19 is the reason one section of TDR-2025-11 concludes narrowly rather than generally.

Reading SEC-2026-05, the defect is that alarm router delivery provider credential shared across regions; the discovery date is 2026-04-16. Assessment finding SEC-2026-05 stands open; committed 2026-09-30 is what the record gives for closure. Severity on SEC-2026-05 is Medium, scored 5.2 by the assessor. After the fix, SEC-2026-05 carries

residual risk described as medium; per-region credentials are scheduled but not issued. Alarm Router finding SEC-2026-05 would be a candidate for the first post-close engineering review under any reasonable plan.

The residual position on SEC-2025-11 is low, stated after mitigation rather than assuming full success. SEC-2025-11 rests on the observation that attribute absence confirmed; no working exploit chain was demonstrated; the reviewer holds that to be strong. The finding at finding SEC-2025-11, dated 2025-09-25, is that console session cookie lacked a strict same-site attribute. SEC-2025-11 against Remote Operations Console produces this judgement: the item is open past its committed date and no revised date has been agreed. assessment finding SEC-2025-11 is reported here in full because a summary of it in an earlier draft of TDR-2025-11 was misread as an assurance.

SEC-2026-04 against WMS Connector matters because customer commercial data is retained in operational logs. Evidence behind assessment finding SEC-2026-04 is that payloads recovered from a staging log store, which the reviewer treats as moderate. For assessment finding SEC-2026-04 the reviewer notes that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. Rating for Medium-severity finding SEC-2026-04 is Medium, band 5.6; the reviewer re-read the scoring rationale and accepted it. WMS Connector finding SEC-2026-04 rests on a single source, and the reviewer has said so rather than implying corroboration.

The reviewer's position on SEC-2026-12 against Edge Gateway Agent is that the finding was closed within the committed window and the closure evidence was inspected. Edge gateway agent retained mutual authentication keys after decommissioning is the substance of Edge Gateway Agent finding SEC-2026-12, raised on 2026-04-20. SEC-2026-12 stands partially remediated; pending is what the record gives for closure. SEC-2026-12 arrives as a Low finding with a 3.5 score, and the reviewer did not adjust either. Edge Gateway Agent finding SEC-2026-12 was cross-checked against the engineering record supplied for TDR-2025-11, and the two agree.

What SEC-2026-20 against Payload Scale Interface enables is this: a calibration change cannot be reconstructed during an investigation. Payload scale calibration record can be edited without an audit entry is the substance of Low-severity finding SEC-2026-20, raised on 2026-04-24. The residual position on finding SEC-2026-20 is low; management accepted pending the calibration service redesign, stated after mitigation rather than assuming full success. The evidence for Payload Scale Interface finding SEC-2026-20 is limited, resting on a configuration read rather than on a demonstrated exploitation: record edited on a laboratory robot. assessment finding SEC-2026-20 has no single accountable owner, and that is the finding the reviewer would press hardest.

Charge Scheduler finding SEC-2025-10 is remediated, with a closure date of 2025-10-09. On 2025-09-24 the tester recorded SEC-2025-10 against Charge Scheduler: charge scheduler exposed an unauthenticated internal health endpoint with site names. After the fix, Charge Scheduler finding SEC-2025-10 carries residual risk described as low. Impact of finding SEC-2025-10 is not theoretical; site naming and fleet size leak to anyone inside the cluster network. Low-severity finding SEC-2025-10 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

As at the review date SEC-2026-02 was remediated and the closure entry reads 2026-05-29. The reason assessment finding SEC-2026-02 was not downgraded is that any cluster workload could read fleet position data for all tenants. The reviewer declines to soften SEC-2026-02 against Robot Shadow Store: the risk acceptance lacks an expiry date, which is a process finding in its own right. High-severity finding SEC-2026-02 reduces to a residual risk of low; authorization is checked on both the cache and durable paths. finding SEC-2026-02 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Schedule L - Vulnerability ageing against the internal remediation window (TDR-2025-11)

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-01	High	22	30 days	yes	network segmentation
SEC-2025-02	High	142	30 days	no	second delivery path
SEC-2025-03	Medium	87	90 days	yes	audit trail only
SEC-2025-04	Medium	43	90 days	yes	none recorded
SEC-2025-05	High	76	30 days	no	restricted role membership
SEC-2025-06	Medium	22	90 days	yes	restricted role membership

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-07	Medium	24	90 days	yes	none recorded
SEC-2025-08	Medium	154	90 days	no	manual review before release
SEC-2025-09	Medium	308	90 days	no	restricted role membership
SEC-2025-10	Low	167	180 days	yes	none recorded
SEC-2025-11	Low	78	180 days	yes	audit trail only
SEC-2025-12	Low	354	180 days	no	restricted role membership
SEC-2026-01	High	232	30 days	no	second delivery path
SEC-2026-02	High	127	30 days	no	manual review before release
SEC-2026-03	High	27	30 days	yes	audit trail only
SEC-2026-04	Medium	135	90 days	no	monitoring alert on use
SEC-2026-05	Medium	100	90 days	no	monitoring alert on use
SEC-2026-06	Medium	192	90 days	no	second delivery path
SEC-2026-07	Medium	158	90 days	no	restricted role membership
SEC-2026-08	Medium	151	90 days	no	monitoring alert on use
SEC-2026-09	Low	150	180 days	yes	manual review before release
SEC-2026-10	Low	78	180 days	yes	monitoring alert on use
SEC-2026-11	Low	200	180 days	no	manual review before release
SEC-2026-12	Low	252	180 days	no	restricted role membership
SEC-2026-13	Low	24	180 days	yes	manual review before release
SEC-2026-14	Low	128	180 days	yes	monitoring alert on use
SEC-2026-15	Low	127	180 days	no	restricted role membership
SEC-2026-16	Medium	148	90 days	no	network segmentation
SEC-2026-17	High	147	30 days	no	second delivery path
SEC-2026-18	Low	305	180 days	no	restricted role membership
SEC-2026-19	Medium	116	90 days	no	network segmentation
SEC-2026-20	Low	97	180 days	no	network segmentation

4. Authentication, authorization and the trust boundaries

Implementation of E-Stop Chain rests on VHDL with discrete relay logic; the process runs on a safety-rated hardware assembly, no general-purpose operating system and persists to no persistent store; events are latched to the safety controller. Supportability of E-Stop Chain turns on one fact: none; the relay parts are current production. On paper E-Stop Chain releases changed only by an engineering change order, and the observed practice did not diverge from that. The objective recorded against E-Stop Chain is that drive power is removed within 250 ms of button actuation on every tested unit, and it is reported weekly to the engineering leadership. Theo Bassett of Safety Engineering maintains E-Stop Chain, and the service inventory entry was current at the review date. E-Stop Chain is the reason one section of TDR-2025-11 concludes narrowly rather than generally.

Automated test coverage on E-Stop Chain stands at no meaningful automated unit coverage, verification being by witnessed test instead. Deployment of E-Stop Chain is edge, hardwired, independent of the compute module, which sets the blast radius of any defect in it. Data ownership is clear for E-Stop Chain: it writes no persistent store; events are latched to the safety controller and no other service writes there. Inputs to E-Stop Chain come from the operator stop buttons and the perimeter interlocks, and its consumers are the safety controller and the drive contactors. Against E-Stop Chain the backlog runs to 2 items, and it includes a known behaviour whereby one pre-production harness revision

remains in the field on nine robots at two sites. E-Stop Chain was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Release cadence for Hardware-in-the-Loop Rig is the harness changes with each firmware release, and the release records bear that out. Hardware-in-the-Loop Rig was assessed on the bench suite definition, rig configuration inventory and two firmware release records, and nothing beyond that was available in the data room. For Hardware-in-the-Loop Rig the reviewer records that the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Hardware-in-the-Loop Rig was checked for shared-database coupling, and the ownership boundary held. Hardware-in-the-Loop Rig's dependency ageing was checked: the rig controller runs an operating system release that left vendor support in 2025. Hardware-in-the-Loop Rig is the clearest instance in TDR-2025-11 of a control that exists on paper and is weaker in practice.

Hardware-in-the-Loop Rig has a single named owner, Nathan Oduya of Robot Firmware, which is not true of every service reviewed. Hardware-in-the-Loop Rig reports 39 per cent line coverage, measured by the same instrumentation the pipeline uses. Inputs to Hardware-in-the-Loop Rig come from the firmware candidate artefacts and the bench sensor stimulators, and its consumers are the firmware release decision record. Hardware-in-the-Loop Rig is deployed laboratory, two rigs, one of which lags the current sensor configuration. Of the 13 open items against Hardware-in-the-Loop Rig, the reviewer treats one as material: the second rig carries a superseded lidar variant, so one sensor path is only tested on one rig. Hardware-in-the-Loop Rig was checked against the company's own register, which had not been updated since the change landed.

Interface discipline around Config Distribution is adequate; the schema is versioned and change is gated on a compatibility check. The objective recorded against Config Distribution is that a configuration change reaches every online robot inside five minutes, and it is reported weekly to the engineering leadership. Documents and data inspected for Config Distribution were the bundle format specification, distribution latency measurements and an offline apply test. Automated test coverage on Config Distribution stands at 72 per cent line coverage. Config Distribution carries 6 open defects, among them a case where a robot that is offline at publication applies the change without re-validating preconditions. Config Distribution is the reason one section of TDR-2025-11 concludes narrowly rather than generally.

The ageing position for Config Distribution is none; the bundle format was revised in 2025, and the reviewer checked it against the vendor statements on file. Config Distribution is owned by the Platform Infrastructure team, with Sam Oyelaran named as the engineering owner of record. Config Distribution sits split edge and cloud, with a signed bundle format, and no alternative placement is currently supported. Config Distribution pairs a Rust 1.79 agent with a Go 1.22 control service implementation with split: an on-robot agent and a cloud service, and its data lives in content-addressed object storage with a PostgreSQL 14 index. Config Distribution was weighed on its coverage, its defect backlog and its dependency position, and the component is adequate today and carries a supportability liability that is not on the funded plan. Config Distribution would be a candidate for the first post-close engineering review under any reasonable plan.

Site Provisioning Service reports 64 per cent line coverage, measured by the same instrumentation the pipeline uses. In production Site Provisioning Service is cloud, global, unchanged since the last architecture review. The backlog for Site Provisioning Service is 5 items deep and contains an acknowledged defect whereby site templates diverged after two customers required bespoke network settings. Where Site Provisioning Service is concerned, the commitment is that a new site reaches a working configuration inside one working day; the reviewer traced the metric to its source. Site Provisioning Service pairs a Go 1.22 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14 with a configuration history table. Site Provisioning Service was reviewed against the position twelve months earlier, and nothing measurable has changed.

Site Provisioning Service sits between the identity service and the customer network intake records and the config distribution service and the WMS connector, and that position is what makes its availability objective load-bearing. For Site Provisioning Service the supportability position is none identified. The Platform Infrastructure team carries Site Provisioning Service on its service inventory, and Marcus Duplessis answered for it in interview. Underlying Site Provisioning Service in this report are the provisioning runbook, template diffs for four sites and two go-live records. The assessment of Site Provisioning Service is not unqualified: it is fit for the current fleet size, and the reviewer found no

evidence that it would remain so at twice that scale. Site Provisioning Service was inspected in the repository rather than described from the ticket, which is why the account is specific.

Safety Controller commits that a validated protective-stop trigger de-energises the drive inside 120 ms, and the objective is measured continuously rather than sampled. Test coverage on Safety Controller is 91 per cent line coverage and the measurement basis was confirmed with Bettina Vogel. The ageing position for Safety Controller is none; the supplier's support commitment runs to 2031, and the reviewer checked it against the vendor statements on file. For Safety Controller the reviewer records that the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. For Safety Controller the reviewer worked from the supplier assessment report, the internal safety case and eight bench trigger tests. Safety Controller is included because a reader of TDR-2025-11 should be able to disagree with the inference without repeating the work.

Safety Engineering releases Safety Controller twice a year, with full revalidation of the safety case, and no release in the sampled window bypassed the gate set. Safety Controller depends on the E-stop chain, the speed and separation monitor and the drive current sensors and is depended on by the drive power contactors and the robot fault annunciator. Safety Controller was built in C, written to a constrained coding subset and has not been rewritten; it still runs on a safety-rated controller module supplied by a third party against a non-volatile fault log on the controller. Safety Controller owns its own data and exposes it only through a versioned interface, which the reviewer read. Against Safety Controller the backlog runs to 3 items, and it includes a known behaviour whereby the fault log wraps after four thousand entries and the oldest records are discarded. Safety Controller was one of two items where TDR-2025-11 could not obtain the underlying data within the engagement.

One coupling concern applies to Analytics Warehouse Loader: a columnar warehouse with daily partitions is shared with a second service, so a schema change is not a local decision. Against Analytics Warehouse Loader the backlog runs to 20 items, and it includes a known behaviour whereby a late-arriving telemetry partition is silently excluded rather than reprocessed. Analytics Warehouse Loader carries a single measurable objective, namely that the daily load completes before 06:00 local time at every region. The manifests place Analytics Warehouse Loader cloud, one loader per region writing to a shared warehouse. Analytics Warehouse Loader was weighed on its coverage, its defect backlog and its dependency position, and the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Analytics Warehouse Loader was cross-checked against the engineering record supplied for TDR-2025-11, and the two agree.

Behind Analytics Warehouse Loader sits a columnar warehouse with daily partitions; the service itself is Python 3.11 with SQL transformations on cloud, a scheduled batch runner. Analytics Warehouse Loader's dependency ageing was checked: one transformation still runs on a deprecated warehouse function. Test coverage on Analytics Warehouse Loader is 45 per cent line coverage and the measurement basis was confirmed with Farida Cisse. Documents and data inspected for Analytics Warehouse Loader were the load schedule, partition completeness checks and two months of load audit records. The customer reporting views and the internal metrics layer rely on Analytics Warehouse Loader, which in turn relies on the telemetry ingest path, the order broker and the billing usage meter. Analytics Warehouse Loader is included because a reader of TDR-2025-11 should be able to disagree with the inference without repeating the work.

Schedule B - Deployment topology and failure domain by service (TDR-2025-11)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Motion Planner	edge, one instance per robot with no cloud fallback	every three weeks, promoted ring by ring	1	one robot	Priya Raghunathan
Costmap Fusion	edge, co-scheduled with the planner on the compute module	monthly, batched with perception model updates	1	one robot	Daniel Okafor
Fleet Allocator	cloud, one regional deployment serving all sites in that region	weekly, behind a change approval	4	one region	Marta Kellerman
Traffic Manager	cloud, regional, with a degraded on-robot yield behaviour if the link drops	weekly, paired with the allocator release	3	one region	Owen Brantley
Charge Scheduler	cloud, regional, with a conservative local	fortnightly, without a formal approval gate	2	one region	Nadia Semenova

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	fallback policy on the robot	until March 2026			
Safety Controller	edge, dual-channel, with cross-checking between channels	twice a year, with full revalidation of the safety case	3	one robot	Bettina Vogel
E-Stop Chain	edge, hardwired, independent of the compute module	changed only by an engineering change order	4	one robot	Theo Bassett
Lidar Driver	edge, one process per sensor	as needed, historically twice a year	3	one robot	Yuki Harada
Localization Engine	edge, with map distribution from the cloud	every three weeks with the planner	6	one robot	Rahul Menon
WMS Connector	cloud, one deployment per customer tenant	every two weeks, per tenant on request	5	one region	Tomas Villareal
Order Broker	cloud, regional	weekly	4	one region	Grace Lindqvist
Telemetry Ingest	cloud, one ingest endpoint per region with an edge buffer on the robot	weekly, with schema changes gated on a compatibility check	2	one robot	Farida Cisse
OTA Updater	split edge and cloud, with staged rings and a per-site pause control	every three weeks, following the autonomy release	4	one robot	Erik Palmgren
Identity and Authorization	cloud, global control plane with regional caches	weekly, with a security review on every change	4	one region	Sonia Krall
Remote Operations Console	cloud, with a thin session relay in each region	twice a week	5	one region	Anjali Sekhon
Simulation Harness	cloud batch, with a local mode for developers	continuous, from the main branch	2	one region	Colin Achebe
Hardware-in-the-Loop Rig	laboratory, two rigs, one of which lags the current sensor configuration	the harness changes with each firmware release	3	one region	Nathan Oduya

4. Authentication, authorization and the trust boundaries - part 2 of 2

Following on directly, these further rows of the 4. Authentication, authorization and the trust boundaries belong to the same schedule and share its numbering. The break is editorial.

Qualifications noted at the head of the 4. Authentication, authorization and the trust boundaries apply to these rows equally, and are not restated for each part. Nothing is restated from the earlier part.

Schedule B - Deployment topology and failure domain by service (TDR-2025-11) (continued, part 2 of 2)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Health Monitor	cloud, regional	fortnightly	2	one region	Wanda Njoku
Battery Management Service	split edge and cloud	pack firmware twice a year, the cloud service monthly	6	one robot	Peter Halvorsen
Gripper Controller	edge, only on the two robot variants fitted with a manipulator	quarterly	2	one robot	Ines Duarte
Map Authoring Studio	cloud, single global deployment	weekly	3	one region	Lucia Ferrante
Site Provisioning Service	cloud, global	fortnightly	4	one region	Marcus Duplessis
Event Bus	cloud, one cluster per region, three availability zones	upgraded twice a year	6	one region	Erik Palmgren
Config Distribution	split edge and cloud, with a signed bundle format	weekly	4	one robot	Sam Oyelaran

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Robot Shadow Store	cloud, regional, sharded by site	weekly	6	one region	Marta Kellerman
Analytics Warehouse Loader	cloud, one loader per region writing to a shared warehouse	weekly	5	one region	Farida Cisse
Billing Usage Meter	cloud, regional	monthly, aligned to the billing cycle	1	one region	Nadia Semenova
Alarm Router	cloud, regional, with a second delivery provider	fortnightly	2	one region	Wanda Njoku
Video Snippet Service	cloud, regional, with an on-robot capture buffer	monthly	5	one region	Anjali Sekhon
Speed and Separation Monitor	edge, cross-checked against the safety controller's own inputs	twice a year, with revalidation	1	one robot	Bettina Vogel
Wheel Odometry Filter	edge	with the firmware, twice a year	1	one robot	Yuki Harada
Payload Scale Interface	edge, only on robots fitted with the weighing deck	as needed	3	one robot	Ines Duarte
Edge Gateway Agent	edge, one per robot, with mutual authentication to the cloud	every three weeks	3	one robot	Sam Oyelaran
Release Orchestrator	cloud, single global deployment	continuous, self-hosting	6	one region	Colin Achebe

5. Supply chain and dependency scanning results

Under Zlib, physics-core-fork is treated as no copyleft in the inventory. Against physics-core-fork the reviewer would add that the fork diverges by nineteen patches and no upstream merge path is maintained. Provenance of physics-core-fork is forked from upstream in 2023 with local patches. physics-core-fork illustrates why TDR-2025-11 separates the severity of a finding from the confidence behind it.

Pgbouncer-managed is licensed under ISC, with copyleft exposure recorded as no. Pgbouncer-managed entered the codebase by this route: public registry, signature verified. Pgbouncer-managed appears at OSS-024 in version 1.21.0; the usage recorded is database connection pooling. pgbouncer-managed is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Version 6.0.0 of media-transcode is the catalogued build at OSS-016, used video snippet transcoding. Media-transcode carries a practical obligation: to share alike any derived asset it distributes. Linking mode matters for media-transcode, which is used video snippet transcoding. media-transcode illustrates why TDR-2025-11 separates the severity of a finding from the confidence behind it.

One point about grid-render is not resolved: the commercial licence removes the copyleft obligation and the reviewer inspected the licence certificate. Beyond licensing, grid-render carries a single-maintainer dependency with no succession plan. Because grid-render is used map authoring geometry rendering, its licence terms reach the delivered software. grid-render was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The licence on lidar-vendor-sdk is commercial, redistributable binary; copyleft exposure is no. Compliance with the lidar-vendor-sdk terms requires to publish the modified files it distributes. Lidar-vendor-sdk carries this qualification: two major versions behind current; the vendor's support statement covers the deployed version to 2027. lidar-vendor-sdk is included because a reader of TDR-2025-11 should be able to disagree with the inference without repeating the work.

No copyleft applies to yaml-parse, whose licence is MIT. At OSS-013 the component is yaml-parse, pinned at 0.15.2 and used configuration parsing on the robot. The open question on yaml-parse is that attribution present. yaml-parse is treated as material to the investment case rather than as a routine engineering item.

Meeting the terms on k8s-operator-framework means nothing beyond retaining the licence text with the source. The inventory entry OSS-028 is k8s-operator-framework 1.31.0, cloud deployment controllers. K8s-operator-framework

carries this qualification: attribution present. k8s-operator-framework sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The supply route for otel-instrument is public registry, signature verified. Under Apache-2.0, otel-instrument is treated as no copyleft in the inventory. The reviewer's note on otel-instrument is that attribution present. otel-instrument is included because a reader of TDR-2025-11 should be able to disagree with the inference without repeating the work.

The inventory entry OSS-017 is wms-adapter-sdk 2.2.0, one customer's warehouse system adapter. Wms-adapter-sdk also carries a component with open advisories that do not affect the deployed configuration, which is a supportability question rather than a licence one. Wms-adapter-sdk is licensed under commercial, per-seat, with copyleft exposure recorded as no. wms-adapter-sdk was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The open question on geom2d-fusion is that dynamic linking preserves the weak-copyleft boundary; counsel confirmed the analysis in 2025. Version 0.9.7 of geom2d-fusion is the catalogued build at OSS-002, used dynamically linked by the fusion pipeline. LGPL-2.1 governs geom2d-fusion, which the inventory marks as weak copyleft. geom2d-fusion sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

On provenance, rtos-shim shows public registry, signature verified. Beyond licensing, rtos-shim carries an upstream project that is active but changes interface frequently. No obligations beyond attribution is the reviewer's observation on rtos-shim. rtos-shim was reviewed against the position twelve months earlier, and nothing measurable has changed.

The open question on protobuf-runtime is that attribution present in both notice files. Protobuf-runtime was obtained as follows: public registry, signature verified. What the company must actually do about protobuf-runtime is to reproduce the notice text in the delivered documentation. protobuf-runtime illustrates why TDR-2025-11 separates the severity of a finding from the confidence behind it.

6. Security incidents and the follow-through on them

The commitment register shows process defect incident INC-2025-146 against an undertaking to fold the reset into the re-run command and make double-counting detectable in the ledger, with Nadia Semenova named and 2025-12-31 committed. Taken with the rest of the record, Larkspur Home incident INC-2025-146 indicates that the incident is unremarkable in itself and revealing about the observability of the surrounding system. Billing Usage Meter incident INC-2025-146 shows a short mitigation and a long tail: not applicable; caught before billing against 11 days. For process defect event INC-2025-146 the follow-through was tested rather than assumed, and landed on 2025-11-30. Process defect incident INC-2025-146 records that on 2025-11-19, a re-run of a failed metering window double-counted usage for one customer. incident INC-2025-146 was excluded from an earlier draft of TDR-2025-11 and reinstated after the reviewer obtained the supporting record.

The post-mortem for incident INC-2025-146 identifies the cause as follows: the re-run procedure required a manual ledger reset that the operator skipped. Larkspur Home incident INC-2025-146 rests on strong on the cause and limited on the contributing factor evidence, which the reviewer states separately from the severity of the finding. So far as reach is concerned, process defect event INC-2025-146 involved no production site and no robot directly, the exposure being to internal systems. Process defect incident INC-2025-146 was detected because the monthly reconciliation caught it before invoicing. In taxonomy terms Larkspur Home incident INC-2025-146 is a process defect incident in Billing Usage Meter. Billing Usage Meter incident INC-2025-146 was raised again with the owning engineer after the initial interview, and the position did not change.

Grading Larkspur Home incident INC-2026-051 on evidence rather than on consequence gives limited, being a single account corroborated only by the ticket timeline. 2 h 18 m elapsed before Larkspur Home incident INC-2026-051 was mitigated, with resolution at 9 days. So far as reach is concerned, Larkspur Home incident INC-2026-051 involved 1 site or tenant with no robot taken out of service. In taxonomy terms Larkspur Home incident INC-2026-051 is a configuration drift incident in Site Provisioning Service. Contributing to Site Provisioning Service incident INC-2026-051, and not repaired by the remediation, is that template drift was not detected because the baseline check compares only the default template. incident INC-2026-051 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The post-mortem for Larkspur Home incident INC-2026-051 identifies the cause as follows: the site template had been forked for a customer network requirement and drifted from the baseline. Larkspur Home incident INC-2026-051 began on 2026-05-06 when a new site went live with a bespoke network template that omitted the outbound certificate pin. What INC-2026-051 tells the reviewer is that the failure was in the detection design rather than in the code that failed. The corrective action for configuration drift incident INC-2026-051 is to check every template against the baseline and re-converge the four forked templates, and 2026-06-15 is the date the team gave. Verification of the Site Provisioning Service incident INC-2026-051 remediation gives the check landed on 2026-05-15; two templates remain forked. INC-2026-051 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Harborline Systems incident INC-2026-068 touched one site and 14 robots, and Harborline Systems was the affected account. The immediate event in Harborline Systems incident INC-2026-068 was that odometry drift exceeded specification on a wet floor and localization confidence fell below the mission threshold, dated 2026-06-16. The classification applied to incident INC-2026-068 is environmental, and the affected component is Wheel Odometry Filter. The corrective action for environmental incident INC-2026-068 is to add a slip detector that flags the condition and issue a site advisory on cleaning windows, and 2026-09-15 is the date the team gave. In environmental event INC-2026-068, the site's cleaning schedule changed and now overlaps the peak shift, which is a process finding rather than a code finding. Wheel Odometry Filter incident INC-2026-068 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The engineering team's root cause for INC-2026-068 is that the filter has no wet-floor slip model and does not flag the condition to the operator, and the reviewer found the analysis sound. Not landed; the detector is prototyped and unvalidated is the position on the INC-2026-068 commitment as at the review date. Harborline Systems incident INC-2026-068 was detected because robots stopped and requested assistance, which the site reported as unplanned downtime. Drawing the evidence in Wheel Odometry Filter incident INC-2026-068 together, the blast radius was set by a deployment topology decision made years earlier. The reviewer grades the environmental event INC-2026-068 evidence as strong, resting on the production traces the team supplied and has not upgraded the finding to compensate. environmental incident INC-2026-068 carries a follow-up request that was satisfied after the draft of TDR-2025-11 was circulated.

Process bypass event INC-2025-131 was detected because a field engineer reported unexpected keep-out zones. The post-mortem for INC-2025-131 identifies the cause as follows: the export function did not check publication state. The affected population in INC-2025-131 was one site and 34 robots. Time to mitigate process bypass incident INC-2025-131 was 3 h 22 m, and time to resolve was 44 days. Beyond the direct cause, Map Authoring Studio incident INC-2025-131 shows that the site had a standing workaround for slow map publication that predated the review step. INC-2025-131 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Process bypass incident INC-2025-131 supports one conclusion and not another: the team's analysis was honest about a gap it could have concealed. Process bypass event INC-2025-131 closure status is the export restriction landed on 2025-11-27; the publication work is open. On 2025-10-14 the record for INC-2025-131 opens: an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. Moderate on the cause and strong on the delivery position, which was checked in the repository is the evidence grade for Map Authoring Studio incident INC-2025-131, and the reader should not read severity into it. The remediation committed after INC-2025-131 was to restrict export to published revisions and shorten the publication path so the workaround is unnecessary, owned by Lucia Ferrante with a due date of 2025-12-15. process bypass event INC-2025-131 is reported at the evidence strength available and has not been promoted to fit the narrative.

Incident INC-2025-152 was contained to no production site and no robot directly, the exposure being to internal systems. INC-2025-152 is recorded as follows on delivery: the rule landed on 2026-01-10; deletion of the historical clips is open. Data retention event INC-2025-152 began on 2025-12-04 when one region retained operator video clips for ninety days against a thirty-day commitment. Detection came through an internal privacy review found the discrepancy, which places data retention event INC-2025-152 on the reported side of the ledger. Mitigation of internal incident INC-2025-152 took not applicable; retention only and full resolution took 37 days. internal incident INC-2025-152 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Incident INC-2025-152 carries a second-order finding: bucket creation was not covered by the configuration baseline check. The reviewer's reading of data retention event INC-2025-152 is that management's severity rating is defensible and its remediation priority is not. Grading INC-2025-152 on evidence rather than on consequence gives strong, resting on the production traces the team supplied. Root cause in incident INC-2025-152 was that the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. In taxonomy terms Video Snippet Service incident INC-2025-152 is a data retention incident in Video Snippet Service. INC-2025-152 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-11.

The reviewer checked the WMS Connector incident INC-2025-058 commitment against the code and release records; landed on 2025-05-30; the backfill was completed on 2025-06-03. Data integrity event INC-2025-058 took 2 h 35 m to stabilise and 21 days to close out. The contributing factor recorded against Dunbar Industrial incident INC-2025-058 is that the integration specification allowed the identifier to be tenant-scoped but did not require it. The commitment register shows Dunbar Industrial incident INC-2025-058 against an undertaking to key suppression on the tenant and identifier pair and backfill the affected order records, with Tomas Villareal named and 2025-07-15 committed. The blast radius of incident INC-2025-058 was 2 site or tenant with no robot taken out of service. incident INC-2025-058 was checked against the company's own register, which had not been updated since the change landed.

WMS Connector accounts for WMS Connector incident INC-2025-058, a data integrity event. The trigger recorded for WMS Connector incident INC-2025-058 on 2025-05-09 was that two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. Data integrity event INC-2025-058 is not read as an isolated failure: the blast radius was set by a deployment topology decision made years earlier. Detection came through the customer reported missing picks the same afternoon, which places Dunbar Industrial incident INC-2025-058 on the reported side of the ledger. Data integrity incident INC-2025-058 traces back to a single fact: duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. incident INC-2025-058 illustrates why TDR-2025-11 separates the severity of a finding from the confidence behind it.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2025-11)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
give every corridor reservation a lease with an expiry and add a site-level queue depth alert	INC-2025-041	Owen Brantley	2025-05-30	landed	interview only
re-derive the time offset on every reconnect and add the affected variant to the second bench ri	INC-2025-047	Yuki Harada	2025-06-15	partly landed	configuration export inspected
release charge windows on a shift-end event and reconcile the calendar nightly	INC-2025-052	Nadia Semenova	2025-06-30	landed	commit inspected
key suppression on the tenant and identifier pair and backfill the affected order records	INC-2025-058	Tomas Villareal	2025-07-15	landed	interview only
make the allocator consult charger availability before excluding a robot	INC-2025-063	Marta Kellerman	2025-07-31	landed	test added and observed
add a leadership and under-replicated partition check to the broker upgrade runbook	INC-2025-071	Erik Palmgren	2025-07-31	landed	runbook change inspected
invalidate the roster cache on a fleet membership event and add a session-level regression test	INC-2025-078	Anjali Sekhon	2025-08-31	landed	configuration export inspected
persist the pause state	INC-2025-084	Erik Palmgren	2025-08-15	landed	runbook change

Commitment	Source	Owner	Committed date	Status	Evidence inspected
and require a design note for any change to the release state machine					inspected
add the connector torque check to incoming inspection and retrofit the affected harness batch	INC-2025-090	Bettina Vogel	2025-09-30	partly landed	interview only
replace the mailbox fallback with a secondary responder and import rotations from the scheduling	INC-2025-096	Wanda Njoku	2025-10-15	partly landed	commit inspected
add a deterministic tie-break and return the pinch-point scenario to the nightly suite	INC-2025-103	Priya Raghunathan	2025-10-31	partly landed	commit inspected
fail the load on an incomplete partition and reprocess the affected days	INC-2025-110	Farida Cisse	2025-10-31	landed	release record inspected
redact authorization headers at the logging layer and cut build log retention to fourteen days	INC-2025-118	Sonia Krall	2025-10-31	partly landed	interview only
annotate staleness on every read path and add a rebalance-under-load test to the platform suite	INC-2025-124	Marta Kellerman	2025-11-30	landed	release record inspected
restrict export to published revisions and shorten the publication path so the workaround is unn	INC-2025-131	Lucia Ferrante	2025-12-15	partly landed	no evidence located
split control traffic onto a dedicated topic with its own quota and revisit the topic inventory	INC-2025-139	Erik Palmgren	2026-01-31	partly landed	interview only
fold the reset into the re-run command and make double-counting detectable in the ledger	INC-2025-146	Nadia Semenova	2025-12-31	landed	runbook change inspected
apply the lifecycle rule through the configuration baseline and delete the over-retained clips	INC-2025-152	Anjali Sekhon	2026-01-31	partly landed	no evidence located

6. Security incidents and the follow-through on them - part 2 of 2

Because the 6. Security incidents and the follow-through on them does not fit one page, its remainder is set out below under the same references. Pagination alone determines where it falls.

Column definitions stated with the first part of the 6. Security incidents and the follow-through on them govern the whole of it, and any qualification recorded there is carried forward here. The break is editorial.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2025-11) (continued, part 2 of 2)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
replace the fallback matcher and re-survey the affected	INC-2026-004	Rahul Menon	2026-06-30	not landed	test added and observed

Commitment	Source	Owner	Committed date	Status	Evidence inspected
aisles					
require a calibration record before returning a manipulator robot to service	INC-2026-009	Ines Duarte	2026-04-15	landed	commit inspected
move precondition validation to apply time and add an offline apply case to the platform suite	INC-2026-015	Sam Oyelaran	2026-03-31	landed	configuration export inspected
pin the full build configuration and add a determinism check to the harness	INC-2026-021	Colin Achebe	2026-05-31	not landed	no evidence located
group on the signal and block closure while a constituent alert remains active	INC-2026-028	Wanda Njoku	2026-04-30	landed	no evidence located
measure closing rate where the sensing permits and re-run the protective distance calculation	INC-2026-033	Bettina Vogel	2026-08-31	not landed	no evidence located
serialise cancellation against assignment and add an hourly orphan sweep	INC-2026-039	Grace Lindqvist	2026-05-31	landed	test added and observed
aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current h	INC-2026-045	Sam Oyelaran	2026-06-30	partly landed	runbook change inspected
check every template against the baseline and re-converge the four forked templates	INC-2026-051	Marcus Duplessis	2026-06-15	partly landed	runbook change inspected
refit the estimator on opportunity-charge data and add an imbalance alert independent of state o	INC-2026-057	Peter Halvorsen	2026-09-30	not landed	no evidence located
require a justification and a second approver within one working day for every emergency release	INC-2026-062	Colin Achebe	2026-08-31	not landed	test added and observed
add a slip detector that flags the condition and issue a site advisory on cleaning windows	INC-2026-068	Yuki Harada	2026-09-15	not landed	release record inspected
Partition the fleet allocator write path	TD-01	engineering leadership	2027-03-31	in design	register entry only
Replace the unmaintained scan-matching library in the localization fallback	TD-02	engineering leadership	2026-12-31	committed	register entry only
Retire the vendored geometry library in costmap fusion	TD-03	engineering leadership	2027-06-30	backlog	register entry only
Raise robot software line coverage above sixty per cent on the perception path	TD-04	engineering leadership	2027-03-31	in progress	register entry only
Bring the second bench rig to the current sensor configuration	TD-05	engineering leadership	2026-09-30	committed	register entry only

Commitment	Source	Owner	Committed date	Status	Evidence inspected
Make the nightly simulation suite blocking	TD-06	engineering leadership	2026-11-30	blocked	register entry only

7. Security debt and the funded remediation plan

Hardware-in-the-Loop Rig item TD-14 appears in the register and is funded from the second quarter onward. About Hardware-in-the-Loop Rig item TD-14 it should be said that requires capital for a third rig rather than engineering effort. Under delivery, entry register item TD-14 is to introduce a second bench rig queue to cut firmware lead time. Hardware-in-the-Loop Rig item TD-14 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Sequencing of resilience item TD-12 matters: it is behind a supplier release outside the company's control. Edge Gateway Agent item TD-12 covers the work to extend the edge buffer to twelve hours. The estimate for register item TD-12 is 1 engineer-quarter and $144,000. TD-12 is the clearest instance in TDR-2025-11 of engineering practice being better than the documentation suggests.

TD-16 on Localization Engine carries an estimate of 1 engineer-quarter; the costed figure is $126,000. On funding, TD-16 on Localization Engine is partly funded, with the second phase unbudgeted. The business consequence recorded against programme item TD-16 is that it a stolen compute module discloses customer warehouse layouts. register item TD-16 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

As at the review date TD-02 was committed, with 2026-12-31 still the stated target. Localization Engine carries Localization Engine item TD-02, the work to replace the unmaintained scan-matching library in the localization fallback. Register item TD-02 cannot start earlier than it does because it is in parallel with the release-process changes. TD-02 was excluded from an earlier draft of TDR-2025-11 and reinstated after the reviewer obtained the supporting record.

Alarm Router item TD-15 is on the register because it one credential compromise affects paging everywhere. The register entry TD-15 reads: add per-region credentials for the paging provider, in the security category. Register item TD-15 is unfunded and dependent on capital rather than engineering effort in the engineering plan the reviewer was shown. register item TD-15 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Exposure carried by Event Bus item TD-09 is that it an unsupported broker underneath every service. Register item TD-09 is committed with a target date of 2027-06-30. Event Bus carries TD-09, the work to upgrade the streaming platform before its support window closes. TD-09 has no single accountable owner, and that is the finding the reviewer would press hardest.

The plan treats TD-13 on Localization Engine as funded and already staffed, which the reviewer checked against the headcount model. Localization Engine item TD-13 carries a 2026-12-31 target and a status of in design. Localization Engine item TD-13 is sequenced behind a customer maintenance window that is not yet agreed. TD-13 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Committed is the status of Hardware-in-the-Loop Rig item TD-05, targeted at 2026-09-30. Hardware-in-the-Loop Rig carries TD-05 on Hardware-in-the-Loop Rig, the work to bring the second bench rig to the current sensor configuration. TD-05 on Hardware-in-the-Loop Rig is on the register because it one sensor path is verified on a single rig. programme item TD-05 rests on a single source, and the reviewer has said so rather than implying corroboration.

8. Findings management accepted or declined to fix

Rating for Medium-severity finding SEC-2026-06 is Medium, band 6.1; the reviewer re-read the scoring rationale and accepted it. Simulation Harness finding SEC-2026-06 records that simulation harness cluster used a long-lived cloud role with broad read access, discovered on 2026-04-17. The reason assessment finding SEC-2026-06 was not downgraded is that a compromised scenario could read production artefact storage. SEC-2026-06 against Simulation Harness reduces to a residual risk of medium; the role was scoped down but still reads one production bucket. finding SEC-2026-06 is the clearest instance in TDR-2025-11 of a control that exists on paper and is weaker in practice.

SEC-2026-09 produces this judgement: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. 2026-04-18 is the discovery date of Fleet Health Monitor finding SEC-2026-09, which finds that fleet health monitor query interface allowed unbounded result sets. SEC-2026-09 reduces to a residual risk of low. At 3.9 the score on Fleet Health Monitor finding SEC-2026-09 places it in the Low group. Fleet Health Monitor finding SEC-2026-09 was cross-checked against the engineering record supplied for TDR-2025-11, and the two agree.

After the fix, High-severity finding SEC-2025-01 carries residual risk described as low once lifetimes are cut to one hour with refresh. SEC-2025-01 against Identity and Authorization was graded High, band 7.4, by Redwater Security engagement RW-2025-3. On 2025-09-16 the tester recorded SEC-2025-01 against Identity and Authorization: service tokens are issued with a twelve-hour lifetime. For SEC-2025-01 against Identity and Authorization the record shows that reproduced by the tester and confirmed in the token audit, which is strong evidence and is graded separately from severity. assessment finding SEC-2025-01 was checked against the company's own register, which had not been updated since the change landed.

2026-04-19 is the discovery date of Billing Usage Meter finding SEC-2026-10, which finds that billing usage ledger export written to a bucket without object lock. The remediation ledger shows SEC-2026-10 against Billing Usage Meter as remediated, 2026-06-05. Residual risk after remediation of assessment finding SEC-2026-10 is low; object lock is applied and verified nightly. Weighing SEC-2026-10 against Billing Usage Meter against the compensating controls, the risk acceptance lacks an expiry date, which is a process finding in its own right. finding SEC-2026-10 would be a candidate for the first post-close engineering review under any reasonable plan.

What Low-severity finding SEC-2025-12 enables is this: a loader defect or compromise could alter source telemetry. On Low-severity finding SEC-2025-12 the tester's evidence was that permissions read from the warehouse grant inventory, and the reviewer grades it moderate, because the demonstration was performed in a laboratory tenant rather than in production. SEC-2025-12 against Analytics Warehouse Loader is rated Low with a numeric band of 4.2 on the tester's scoring scale. Analytics warehouse service account held write access to raw telemetry is the substance of Low-severity finding SEC-2025-12, raised on 2025-09-25. Low-severity finding SEC-2025-12 is the entry that changed most between the first and second document requests behind TDR-2025-11.

On 2025-09-18 the tester recorded assessment finding SEC-2025-04: map authoring export bypasses the publication review. Residual risk after remediation of Medium-severity finding SEC-2025-04 is medium; the shortened publication path is still outstanding. On finding SEC-2025-04 the tester's evidence was that reproduced, and independently observed in a field incident, and the reviewer grades it strong. Assessment finding SEC-2025-04 is rated Medium with a numeric band of 5.4 on the tester's scoring scale. finding SEC-2025-04 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Assessment finding SEC-2025-08 is remediated, with a closure date of 2025-10-31. SEC-2025-08 is supported by the fact that reachability proven by the tester from a deployed container; evidence strength is moderate. Event Bus finding SEC-2025-08 reduces to a residual risk of low; the interface is restricted to the operations subnet. On finding SEC-2025-08 the conclusion is that the item is open past its committed date and no revised date has been agreed. SEC-2025-08 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

At 8.1 the score on High-severity finding SEC-2025-02 places it in the High group. In assessment finding SEC-2025-02 the exploitation path means any engineer with build log access could replay a service credential. Assessment finding SEC-2025-02 is not fully retired by its remediation: low; redaction is enforced in the logging library. Tracking High-severity finding SEC-2025-02 to closure gives a status of remediated and a date of 2025-09-19. SEC-2025-02 has no single accountable owner, and that is the finding the reviewer would press hardest.

9. Where a security failure would become a safety failure

Field history for Occasional-exposure hazard HZ-22 is that it has never activated in the field and is evidenced only on the bench. Equipment damage hazard HZ-22 remains Tolerable for outages under the tested duration after the control is applied, and the register marks it partially verified. The parameter set for hazard HZ-22 reads Equipment damage, Occasional, Medium. hazard HZ-22 in the register was tested rather than accepted, and the test is described in the working papers behind TDR-2025-11.

The verification method for Improbable-exposure hazard HZ-11 is supplier abuse test report plus a review of pack fault records. The record shows that hazard HZ-11 in the register produced one field event that did not result in harm. Improbable-exposure hazard HZ-11 scores Fire on severity; exposure is Improbable and controllability Low. Fire hazard HZ-11 rests on a single source, and the reviewer has said so rather than implying corroboration.

Serious injury harm, Occasional exposure and Medium controllability are the parameters recorded for hazard HZ-03. On history, HZ-03 produced no event, and the reviewer regards the absence as weak evidence given the exposure period. Exposure under register entry HZ-03 extends to every robot variant currently in production. Occasional-exposure hazard HZ-03 would move from a concern to a verified defect if one further measurement were supplied.

Against register entry HZ-17 the team relies on periodic fault log extraction to the cloud with a retention policy. Where hazard HZ-17 in the register bites is the newest three sites, where commissioning is incomplete. Verification of No direct harm hazard HZ-17 was by log extraction record review across the fleet. HZ-17 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The three risk terms on hazard HZ-14 in the register are severity Minor injury, exposure Occasional and controllability Low. The residual entry for hazard HZ-14 in the register reads Tolerable once the stale roster defect is closed, verified. HZ-14 produced two near-miss reports from site staff, which is the strongest available test of the control. Minor injury hazard HZ-14 is the clearest instance in TDR-2025-11 of engineering practice being better than the documentation suggests.

The population exposed to register entry HZ-21 is every robot variant currently in production. Risk parameters for HZ-21 are Minor injury severity, Remote exposure and Low controllability. Remote-exposure hazard HZ-21 has produced no field event in the reviewed period; that is worth stating alongside the assessed exposure. HZ-21 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

HZ-19 depends for its mitigation on dock occupancy sensing plus a low approach speed inside the dock envelope. The hazard recorded at hazard HZ-19 in the register is that robot enters a dock occupied by a person. Hazard HZ-19 in the register was scoped to the two pallet-mover variants only, and the reviewer checked the scoping against the fleet record. HZ-19 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Field history for hazard HZ-13 is that it has produced no event, though the control has activated repeatedly. Where register entry HZ-13 bites is every site with a mezzanine or ramp. For register entry HZ-13 the verifying activity was obstacle detection bench test plus a site procedure review. hazard HZ-13 in the register is the reason one section of TDR-2025-11 concludes narrowly rather than generally.

Register entry HZ-02 has produced no field event in the reviewed period, and the register was not updated to reflect it. The design measure recorded at hazard HZ-02 is aisle-end speed reduction plus a mirrored sensing field at mapped blind corners. Tolerable at commissioned sites; two sites have unmapped corners is the residual position on register entry HZ-02, whose status is partially verified. Occasional-exposure hazard HZ-02 was one of two items where TDR-2025-11 could not obtain the underlying data within the engagement.

10. Open findings, risk acceptances and qualifications

The open items in this report are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. J. Ferreiro has dated each open item in TDR-2025-11 so that a reader can see which have been open longest rather than only how many there are. For the transaction, the present report identifies 1 matters the reviewer would raise before closing and 3 for the post-close engineering plan. Deal implications in TDR-2025-11 are modest in aggregate and specific in detail: 1 items for the closing checklist, 3 for the plan. This report does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

TDR-2025-11 keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. Where management has declined to act, report TDR-2025-11 preserves the decision and the stated rationale so a post-close plan can revisit it. The present report is limited by log retention: the telemetry window is ninety days, so events before 2026-02-20 could not be examined directly. The report at TDR-2025-11 carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent

measurement inside the engagement. Report TDR-2025-11 was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

Nothing in this report is presented as verified on the strength of an interview alone. Readers of the report at TDR-2025-11 should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. Wrenfield Technical Advisory LLP asked the same four questions of every team interviewed for the present report, which is how the inconsistencies below were found. Minutes of every report TDR-2025-11 interview were returned to the participant, and three came back with substantive corrections. The present report is one input to a diligence process and is not a substitute for the company's own engineering judgement.

The report at TDR-2025-11 proposes 4 metrics for the post-close monitoring pack, each already produced by an existing system. Wrenfield Technical Advisory LLP recommends that the 4 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. J. Ferreiro tested a sample rather than the population in TDR-2025-11, and the sampling basis is stated in each section. J. Ferreiro was granted read access to the code, the pipeline and the incident tracker for the purposes of this report; production access was not requested. Wrenfield Technical Advisory LLP has costed remediation at the company's own loaded engineering rate rather than at a market rate.

11. Conclusions on the security position

For Ardent Peak's purposes, the present report concludes that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. Stated without qualification for once, the conclusion of report TDR-2025-11 is that the technical estate would support the business plan through the next twelve months and not through the fleet size the plan assumes for 2027 without the identified programme. TDR-2025-11 records 2 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. One declinature in TDR-2025-11 rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. Wrenfield Technical Advisory LLP was given read access without restriction on scope within the agreed subject matter.

None of the findings in the present report would in the reviewer's view justify withholding completion, and two would justify a specific undertaking. The present report suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. J. Ferreiro interviewed 18 engineers and managers for TDR-2025-11; two declined to be quoted on one topic and that is noted where it matters. 15 follow-up items arose from the interview programme for the report at TDR-2025-11; 4 were still open at delivery. J. Ferreiro interviewed 18 people for report TDR-2025-11; where an account was uncorroborated it is marked as such.

Report TDR-2025-11 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. Monitoring recommended by the report at TDR-2025-11 relies on metrics the company already emits, so no new instrumentation is required to start. 7 sites contributed evidence to this report, and one further site was in acceptance testing and was excluded. J. Ferreiro chose the sample for the present report after reading the dependency map, so the selection is not proportionate to headcount or to code volume. Report TDR-2025-11 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.